

hw5

Problem 1

Let $A = L_1 U_1$ and $A = L_2 U_2$, then $L_1 U_1 = L_2 U_2$

since we know that A is invertible, L_1, L_2, U_2, U_1 are invertible so this implies $U_1 U_2^{-1} = L_1^{-1} L_2$

Since U_2 is upper triangular, so is U_2^{-1} and U_1 is upper triangular so then $U_1 U_2^{-1}$ must be upper triangular

Likewise, $L_1^{-1} L_2$ is lower triangular

So a lower triangular matrix is equal to an upper triangular matrix which is only possible if they are diagonal matrices. So let this matrix be D

Since L_1 and L_2 is unit triangular, so is L_2^{-1} so $L_1^{-1} L_2$ is also unit triangular therefore $D = I_m$ so

$$U_1 U_2^{-1} = I_m \implies U_1 = U_2$$

and

$$L_1^{-1} L_2 = I_m \implies L_1 = L_2$$

So LU factorization is unique

Problem 2

Proof by contradiction:

Let A be nonsingular and for some k, $a_{ik}^{(k)}$ where $i = k \dots m$ are all 0

If all $a_{ik}^{(k)} = 0$, this means that $a_{kk}^{(k)} = \dots = a_{mk}^{(k)}$ is equal to 0.

Since we are doing GEPP, at step k, $a_{1,k-1}^{(k)} \dots a_{k-1,k-1}^{(k)} \neq 0$ but $a_{k,k-1}^{(k)} \dots a_{m,k-1}^{(k)} = 0$,

Likewise, $a_{1,k}^{(k)} \dots a_{k-1,k}^{(k)} \neq 0$ and we are given that $a_{k,k}^{(k)} \dots a_{m,k}^{(k)} = 0$.

From here we can see that the columns of $A^{(k)}$ are linearly dependent (they have the same column span) so $A^{(k)}$ must be linearly dependent

Since GEPP is the same as multiplying elementary matrices to a matrix, this means that there is some nonsingular matrix $P^{(k)}$ that is the product of elementary matrices st $P^{(k)} A = A^{(k)}$.

Since we know P and A are nonsingular ($\det \neq 0$), $\det(A^{(k)}) \neq 0$ so $A^{(k)}$ is nonsingular which is a contradiction

Problem 3

a)

Problem 3. A matrix $B \in \mathbb{R}^{n \times n}$ is said to be strictly column diagonally dominant (scdd) if

$$|b_{jj}| > \sum_{i=1, i \neq j}^n |b_{ij}|, \quad j = 1, \dots, n.$$

(a) Show that, in the exact arithmetic, GE without pivoting preserves scdd property, that is, show that if GE without pivoting is applied to a scdd $A \in \mathbb{R}^{m \times m}$, then for each $k = 1, \dots, m$, the subblock of $A^{(k)}$ obtained by deleting from $A^{(k)}$ the first $k-1$ rows and the first $k-1$ columns is also scdd. **Hint:** Use induction on k and consider step k for obtaining $A^{(k+1)}$ from $A^{(k)}$. Also you may want to use the inequalities $|\alpha \pm \beta| \leq |\alpha| + |\beta|$, $|\alpha| - |\beta| \leq |\alpha - \beta|$ that hold for any $\alpha, \beta \in \mathbb{R}$.

Base case: at $A^{(1)} = A$ and we know A is scdd so we are done

Inductive hypothesis: Assume $A_{(k:m, k:m)}^{(k)}$ is scdd

Proving for $k+1$:

Know that at step $k+1$, any element element of A is the result of gaussian elimination so we have:

$$a_{ij}^{(k+1)} = a_{ij}^{(k)} - \frac{a_{ik}^{(k)}}{a_{kk}^{(k)}} a_{kj}^{(k)}$$

So then we want to prove the equation in the question. The RHS is

$$\sum_{\substack{i=k+1 \\ i \neq j}}^m |a_{ij}^{(k+1)}| = \sum_{\substack{i=k+1 \\ i \neq j}}^m \left| a_{ij}^{(k)} - \frac{a_{ik}^{(k)}}{a_{kk}^{(k)}} a_{kj}^{(k)} \right| < \sum_{\substack{i=k+1 \\ i \neq j}}^m |a_{ij}^{(k)}| + \frac{|a_{kj}^{(k)}|}{|a_{kk}^{(k)}|} \sum_{\substack{i=k+1 \\ i \neq j}}^m |a_{ik}^{(k)}|$$

and by the inductive hypothesis we know $\sum_{\substack{i=k+1 \\ i \neq j}}^m |a_{ij}^{(k)}| + |a_{kj}^{(k)}| < |a_{jj}^{(k)}|$ (elem a_{kj} is not included in the sum)

$$\text{so } \sum_{\substack{i=k+1 \\ i \neq j}}^m |a_{ij}^{(k)}| < |a_{jj}^{(k)}| - |a_{kj}^{(k)}|$$

$$< |a_{jj}^{(k)}| - |a_{kj}^{(k)}| + \frac{|a_{kj}^{(k)}|}{|a_{kk}^{(k)}|} \sum_{\substack{i=k+1 \\ i \neq j}}^m |a_{ik}^{(k)}|$$

From here we know that $\sum_{\substack{i=k+1 \\ i \neq j}}^m |a_{ik}^{(k)}| + |a_{jk}^{(k)}| < |a_{kk}^{(k)}|$ (from the induct hypthothesis as well since a_{jk} isnt included in the sum)

$$\text{so } \sum_{\substack{i=k+1 \\ i \neq j}}^m |a_{ik}^{(k)}| < |a_{kk}^{(k)}| - |a_{jk}^{(k)}|$$

$$< |a_{jj}^{(k)}| - |a_{kj}^{(k)}| + \frac{|a_{kj}^{(k)}|}{|a_{kk}^{(k)}|} (|a_{kk}^{(k)}| - |a_{jk}^{(k)}|) = |a_{jj}^{(k)}| - \frac{|a_{kj}^{(k)}|}{|a_{kk}^{(k)}|} |a_{jk}^{(k)}|$$

And the LHS is

$$|a_{jj}^{(k+1)}| = \left| a_{jj}^{(k)} - \frac{a_{jk}^{(k)}}{a_{kk}^{(k)}} a_{kj}^{(k)} \right| \geq |a_{jj}^{(k)}| - \frac{|a_{jk}^{(k)}|}{|a_{kk}^{(k)}|} |a_{kj}^{(k)}|$$

So now we can see that $|a_{jj}^{(k+1)}| \geq |a_{jj}^{(k)}| - \frac{|a_{jk}^{(k)}|}{|a_{kk}^{(k)}|} |a_{kj}^{(k)}| > \sum_{i=k+1}^m |a_{ij}^{(k+1)}|$

So we have proven for the case of $k+1$ so we are done

b)

Proving SCDD never fails:

From the last problem we know that all steps of a scdd matrix produce a scdd submatrix

So at step k , we know the pivot is $a_{kk}^{(k)}$ and from defn, we know $|a_{kk}^{(k)}| > 0$ (since the smallest the column non diagonal elements can be is 0 and the diagonal element is strictly bigger than 0, so $a_{kk}^{(k)} > 0$)

Since $a_{kk}^{(k)} > 0$, we can use GE so at each step k , GE will work so GE works for an SCDD matrix

Showing $|l_{ik}| < 1$:

Since $|l_{ik}| = \left| \frac{a_{ik}}{a_{kk}} \right|$ and we know that $|a_{kk}| > |a_{ik}|$ by properties of SCDD (because if $a_{ik} = x$ and all other elems are 0, then $a_{kk} > x$)

So $|l_{ik}| = \left| \frac{a_{ik}}{a_{kk}} \right| < 1$ so we are done!

Problem 4

```
function x = forsub(L, b)
    n = length(b);
    x = zeros(n, 1);
    x(1) = b(1) / L(1,1);
    for i = 2:n
        x(i) = (b(i) - L(i, 1:i-1) * x(1:i-1)) / L(i,i);
    end
end

function x = bacsub(U, b)
    n = length(b);
    x = zeros(n, 1);
    x(n) = b(n) / U(n,n);
    for i = n-1:-1:1
        x(i) = (b(i) - U(i, i+1:n) * x(i+1:n)) / U(i,i);
    end
end
```

```

A = pascal(12)
x = ones(12, 1);
b = A * x

[L, U, P] = lu(A);
b_new = P * b;

y = forsub(L, b_new);
x_bar = bacsu(U, y);

err = norm(x_bar-x)/norm(x)
bound = eps * cond(A)

```

Output:

A =

Columns 1 through 6

1	1	1	1	1	1
1	2	3	4	5	6
1	3	6	10	15	21
1	4	10	20	35	56
1	5	15	35	70	126
1	6	21	56	126	252
1	7	28	84	210	462
1	8	36	120	330	792
1	9	45	165	495	1287
1	10	55	220	715	2002
1	11	66	286	1001	3003
1	12	78	364	1365	4368

Columns 7 through 12

1	1	1	1	1	1
7	8	9	10	11	12
28	36	45	55	66	78

84	120	165	220	286	364
210	330	495	715	1001	1365
462	792	1287	2002	3003	4368
924	1716	3003	5005	8008	12376
1716	3432	6435	11440	19448	31824
3003	6435	12870	24310	43758	75582
5005	11440	24310	48620	92378	167960
8008	19448	43758	92378	184756	352716
12376	31824	75582	167960	352716	705432

b =

12
78
364
1365
4368
12376
31824
75582
167960
352716
705432
1352078

err =

7.9681e-06

bound =

1.9460e-04

A and b are floating point matrices and vectors because both of them can be represented by floating point numbers accurately. The minimum positive value in a Matlab double is $\approx 10^{-308}$ which is well under the minimum value of our matrix/vector, 1. The maximum number is $\approx 10^{308}$

which is well over the maximum value of our matrix/vector, 1352078. Additionally since there are no floating point numbers, each number in the matrix/vector can be represented perfectly by Matlab so they are floating point matrices and vectors.

We can also see the error is less than our bound

Problem 5

Know $Ax = b$ and $A = A_1 A_2$ and $(A_1 + \delta A_1)\hat{y} = b$ and $(A_2 + \delta A_2)\tilde{x} = \tilde{y}$

$$\text{so } (A_1 + \delta A_1)(A_2 + \delta A_2)\tilde{x} = b$$

$$\text{so } (A_1 A_2 + A_1 \delta A_2 + \delta A_1 A_2 + \delta A_1 \delta A_2)\hat{x} = b$$

$$\text{Since } \|A\| = \|A_1 A_2\| \leq \|A_1\| \|A_2\|, \text{ we have } \delta A = A_1 \delta A_2 + \delta A_1 A_2 + \delta A_1 \delta A_2$$

$$\text{because then } (A + \delta A)\hat{x} = b$$

So by triangle inequality

$$\|\delta A\| = \|A_1 \delta A_2 + \delta A_1 A_2 + \delta A_1 \delta A_2\| \leq \|A_1 \delta A_2\| + \|\delta A_1 A_2\| + \|\delta A_1 \delta A_2\|$$

and

$$\|A_1 \delta A_2\| \leq \|A_1\| \|\delta A_2\| \leq c(m)u \|A_1\| \|A_2\|$$

$$\|\delta A_1 A_2\| \leq \|\delta A_1\| \|A_2\| \leq c(m)u \|A_1\| \|A_2\|$$

$$\|\delta A_1 \delta A_2\| \leq (c(m)u)^2 \|A_1\| \|A_2\|$$

so then

$$\|\delta A\| \leq (2c(m)u + c(m)^2 u^2) \|A_1\| \|A_2\|$$

since $c^2(m)u^2$ is negligible compared to $2c(m)u$, we get

$$\|\delta A\| \leq 2c(m)u \|A_1\| \|A_2\|$$

For it to be backward stable, we want it in the form $\|\delta A\| \leq Cu\|A\|$

so if $\|A_1\| \|A_2\| \leq K\|A\|$ then $\|\delta A\| \leq 2c(m)u \|A\| K$

So we get

$$\|A_1\| \|A_2\| \leq K\|A\| \quad \text{where } K \text{ not too large}$$

$$\text{or equivalently } \frac{\|\delta A\|}{\|A\|} \leq 2c(m)u \left(\frac{\|A_1\| \|A_2\|}{\|A\|} \right)$$

So we want $\frac{\|A_1\| \|A_2\|}{\|A\|}$ to be sufficiently small, probably $O(1)$

Same thing but written:

$Ax = \tilde{b} \quad A = A_1 A_2 \quad w/ \quad (A_1 + \delta A_1) \tilde{y} = \tilde{b} \quad + \quad (A_2 + \delta A_2) \tilde{x} = \tilde{y}$
 then $(A_1 + \delta A_1)(A_2 + \delta A_2) \tilde{x} = \tilde{b}$ so
 $(A_1 A_2 + A_1 \delta A_2 + \delta A_1 A_2 + \delta A_1 \delta A_2) \tilde{x} = \tilde{b}$
 Know $\|A\| = \|A_1 A_2\| \leq \|A_1\| \|A_2\|$
 so $\delta A = A_1 \delta A_2 + \delta A_1 A_2 + \delta A_1 \delta A_2$ b/c then
 $(A_1 + \delta A) \tilde{x} = \tilde{b}$
 $\|\delta A\| = \|A_1 \delta A_2 + \delta A_1 A_2 + \delta A_1 \delta A_2\| \leq \|A_1 \delta A_2\| + \|\delta A_1 A_2\| + \|\delta A_1 \delta A_2\|$
 $\|A_1 \delta A_2\| \leq \|A_1\| \|\delta A_2\| \leq \|A_1\| c(m) u \|A_2\| = c(m) u \|A_1\| \|A_2\|$ (call this $\downarrow$)
 $\|\delta A_1 A_2\| \leq \|\delta A_1\| \|A_2\| \leq c(m) u \|A_1\| \|A_2\|$
 $\|\delta A_1 \delta A_2\| \leq (c(m) u)^2 \|A_1\| \|A_2\|$
 so then $\|\delta A\| \leq (2c(m)u + c(m)^2 u^2) (\|A_1\| \|A_2\|)$ since $c^2(m)u^2$ is negligible compared to $2c(m)u$
 $\approx 2c(m)u \|A_1\| \|A_2\|$ so $\|\delta A\| \leq 2c(m)u \|A_1\| \|A_2\|$
 for it to be br stable we need $\|\delta A\| \leq \epsilon$
 want in the form $\|\delta A\| \leq c u \|A\|$
 so if $\|A_1\| \|A_2\| \leq K \|A\|$ then $\|\delta A\| \leq 2c(m)u \|A\| \cdot K$
 so we want K not too big st $\|A_1\| \|A_2\| \leq K \|A\|$ OR
 $\frac{\|\delta A\|}{\|A\|} \leq \frac{2c(m)u \|A_1\| \|A_2\|}{\|A\|}$ so want $\frac{\|A_1\| \|A_2\|}{\|A\|}$ to be sufficiently small
 $\alpha(1)$ probably

Problem 6

Let $A \in \mathbb{R}^{m \times m}$ be symmetric and positive definite

a)

Let $X \in \mathbb{R}^{m \times m}$ be nonsingular and $X^T A X = B$

then $B^T = (X^T A X)^T = X^T A^T X = X^T A X = B$ so B is symmetric

Let $x \in \mathbb{R}^m$, where x is not the 0 vector

let $Xx = y$, y cannot be the 0 vector because X is nonsingular and $x^T X^T = (Xx)^T = y^T$

Then $x^T B x = x^T X^T A X x = y^T A y$

Since $y \in \mathbb{R}^m$ and A is positive definite, then $y^T A y > 0$ so
 $x^T B x > 0 \forall x \neq 0$

b)

Let $B = A(j:k, j:k)$ where $1 \leq j \leq k \leq m$. Since $A^T = A$, know that $A_{f,g} = A_{g,f} \forall f, g \leq m$ so then for all elems in B, $B_{h,i} = A_{j+h,j+i} = A_{j+i,j+h} = B_{i,h}$ so then $B = B^T$ so it is symmetric

Since A is positive definite, we know $\forall x \neq 0 \in \mathbb{R}^m, Ax \neq 0$

So pick x st $x_1 = \dots = x_{j-1} = x_{k+1} = \dots = x_m = 0$ and $x_i \neq 0$ where $j \leq i \leq k$, then

and call $y = [x_j \dots x_k]^T$

$x^T A x$ is equivalent to $y^T B y$ since x^T eliminates all of the rows not in B and x eliminates all the cols not in B

And since $x^T A x > 0$, $y^T B y > 0$ so B is positive definite